

SE – Lab 4

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Use-Case Flow Specification

Problem Statement #51 – Virtual Technical Conference Platform

Use Case ID: UC-05

Use Case Name: Generate Conference Schedule

Primary Actor: Track Chair

1. Preconditions

1. Accepted papers/presentation sessions are available in the system.
2. Conference tracks have been defined.
3. Available rooms and their seating capacities have been recorded.
4. Speaker availability information has been provided.
5. Required scheduling constraints are available to the system.

2. Main Success Scenario

1. The **Track Chair** selects "**Generate Conference Schedule**".
2. The system retrieves the accepted presentation sessions.
3. The system retrieves the available conference tracks and rooms.
4. The system retrieves the availability of the speakers.
5. The system checks room capacities against the requirements of each presentation session.
6. The system checks speaker availability for the available time slots.
7. The system assigns each presentation to a suitable track, room, and time slot.
8. The system verifies that all room-capacity and speaker-availability constraints are satisfied.
9. The system generates the conflict-free multi-track conference schedule.

10. The system displays the generated schedule to the Track Chair.
11. The Track Chair reviews and approves the schedule.
12. The system publishes the approved schedule.

3. Alternate Flow – Scheduling Conflict

3a. Room Capacity or Speaker Availability Conflict

1. During schedule generation, the system detects that a proposed session assignment violates a scheduling constraint.
2. The system identifies the conflicting session and the violated constraint.
3. The system rejects the conflicting assignment.
4. The system searches for another suitable room or available time slot.
5. **If a valid alternative is found**, the system assigns the session to the alternative and continues with schedule validation.
6. **If no valid alternative is found**, the system reports the unscheduled session to the Track Chair.
7. The Track Chair modifies the scheduling inputs or available constraints.
8. The system attempts to generate the schedule again.

4. Postconditions

If Successful

- A valid multi-track conference schedule has been generated.
- All scheduled sessions satisfy room-capacity constraints.
- Speakers are scheduled only during their available time slots.
- The approved schedule is published for conference users.

If Unsuccessful

- The conflicting or unscheduled sessions are identified.
- An invalid schedule is not published.